Dynamic Driving Risk Potential Field Model Under the Connected and Automated Vehicles Environment and Its Application in Car-Following Modeling

Linheng Li, Jing Gan, Xinkai Ji, Xu Qu, and Bin Ran

***Abstract*— This paper proposes a new dynamic driving risk potential field model under the connected and automated vehicles environment that fully considers the dynamic effect of the vehicle's acceleration and steering angle. The statistical analysis of the model's parameter reveals that acceleration and steering angle will directly affect the distribution of the driving risk potential field and that this strong correlation should not be ignored if one is interested in the vehicle's microscopic motion behavior. We further develop a driving risk potential field-based car-following model (DRPFM) to remedy the failure of acceleration consideration under the conventional environment, whose parameters are calibrated by filtered I-80 NGSIM data with frequent traf?c oscillations. Simulation results indicate that our proposed DRPFM model is proved to be a good description of car-following behavior and outperforms two classical car-following models (Optimal Velocity Model and Intelligent Driver Model) in frequent oscillation phases due to our consideration of potential acceleration data acquisition in real-time under the CAVs environment. In addition, this DRPFM model is applied to deduce the safety conditions for vehicle lane-changing. The analysis results prove that this model can reasonably explain the influencing factors between driver types and lane-changing safety conditions in practice.**

***Index Terms*— Driving risk potential field, car-following model, lane-changing model, connected and automated vehicle system.**

I. Introduction

THE application of potential field theory in traffic flow modeling has existed for a long time. At the beginning of the 21st century, some scholars put forward the concept of the potential field and applied it to robot path planning. Inspired by this idea, many scholars extended the potential field theory to the field of traffic flow research and obtained some research results. Wolf and Burdick established the corresponding potential field model for different objects, creatively constructed the vehicle potential field into a wedge form, and established the mapping relationship between the potential field and traffic behavior [1]. Ni *et al.* demonstrated the objectivity and universality of the potential field in traffic from both macro and micro perspectives, and calibrated the potential field car-following model using NGSIM (Next Generation Simulation) data [2]–[4]. Li *et al.* proposed a simple car-following model based on the concept of the potential field from the perspective of stimulus-response, but the factors considered in the model are relatively simple and the potential field model established is relatively simple [5]. Wang *et al.* established a unified model to characterize the "driving risk field" based on the previous research, put forward the concept of "driving risk field", and validated the model with real vehicles [6], [7]. The results show that the model can provide an effective method for evaluating the driving risk in a complex traffic environment. Based on game theory and traffic field theory Li *et al.* established a safety control model to evaluate the driver's driving risk [8]. It is worth mentioning that by improving the potential field model proposed by Wang et al, they optimized the vehicle's driving field into an elliptic structure, which is more realistic in model interpretation. In summary, the potential field theory can describe the interrelationship between various factors in a complex traffic environment and can explain various phenomena in the real traffic environment with a unified framework.

In recent years, the Connected and Automated Vehicles (CAVs) system has achieved unprecedented development. The CAVs system is the third development stage of the Intelligent Transportation System (ITS) after dynamic perception and active management [9]. Benefited by enormously advanced communication technologies, a variety of vehicle motion information (e.g., position, velocity, and acceleration) can be transmitted and shared in real-time, vehicle to vehicle, as well as vehicle to infrastructure. Under the CAVs system, vehicles are capable of receiving accurate information about other traffic subjects, such as motion information of the surrounding vehicles, the position of the pedestrians on the crossroad, even information on the signal light ahead. Vehicles can thus make efficient operating decisions by synthesizing such information, and the transportation system will eventually become intelligent, networked and integrated. Based on the above characteristics of the CAVs system, traffic operation efficiency, traffic congestion, road capacity, energy consumption, and safety performance can be greatly improved under this environment [10]–[13]. Additionally, in bringing

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these improvements, the arrival of CAVs is bound to influence traffic flow characteristics. Under the traditional traffic environment, human-driven vehicles can only receive very limited motion information, most of which is inaccurate, artificially perceived, and with fuzzy values, such as the distance of the surrounding vehicles, the velocity of the leading vehicle, and the velocity of the object vehicle. This fuzzy numerical feedback and the variations in feedback from different drivers will result in different driving behaviors. Fortunately, in the CAVs environment, vehicles can receive a variety of motion information, the values of which are not only accurate, but most importantly, some motion information that cannot be directly perceived by human drivers could be perceived by the vehicles, for example (acceleration and the steering angle of surrounding vehicles). Therefore, vehicles receive multiple motion information and can provide feedback on the required motion behavior in combination with the actual traffic environment and driving requirements.

Microscopic models have always been a hot spot in traffic flow research. Car-following and lane-changing are the basic microscopic behaviors of traffic flow. The microscopic model of car-following and lane-changing plays the role of a bridge that connects microscopic driving behavior and macroscopic traffic flow characteristics. Various car-following and lane-changing models have been proposed and upgraded in order to be more practical with the real traffic condition. In recent years, more and more attention has been paid to the research of car-following and lane-changing models under the CAVs system. See for example [14], [15] for reviews. Although different microscopic models have different structures and expressions due to different perspectives of modeling, strict boundaries don't exist between these models. All of these models are based on the actual operation of vehicles, in order to effectively characterize traffic flow and microscopic behavior characteristics through analysis of the models. The difference is only that the methodologies used are slightly different, due primarily to the fact that transportation is cross-disciplinary and different theories need to be exploited in order to conduct research. In 1986, the concept of an artificial potential field was first postulated and applied to robot path planning [16]. Inspired by this idea, many scholars extended the potential field theory to the field of traffic flow research and obtained some important research results [1], [2], [6], [17]. It can be found that in the traditional environment, the potential field theory is a rarely used method, as although it can combine many traffic factors for theoretical modeling in the complex traffic environment, there is a fatal flaw in the potential field model established in the traditional environment as described above. This directly leads to the fact that the potential field model established in the traditional environment can only characterize the variation of the potential field at a certain spatial position and the current fixed velocity. We can call these models "static models". Due to the lack of an acceleration parameter, the distribution of the potential field cannot reflect the change of the object's motion trend, so it cannot reflect the dynamic change of the potential field with the change of vehicle motion states. However, some key vehicle motion information (such as acceleration of the surrounding vehicles) which couldn't be obtained from the traditional traffic environment can be completely obtained under the CAVs environment, but. practical application of these current potential field models under the CAVs system is limited. The current potential field theory, for the most part, has been used for the description of traffic behavior. The theory has not been widely used in microscopic traffic flow modeling.

Therefore, this paper aims to construct a dynamic driving risk potential field model and describes various microscopic driving behaviors in detail through the model. The proposed dynamic driving risk potential field in this paper can characterize the degree of safety risk during the driving process. The value of the driving risk potential field directly affects the microscopic driving behavior of vehicles. The microscopic behaviors can be treated as a result of the safety effect due to the fact that the driving safety conditions matter when a driver is going to take their next action. For example, most of the reasons vehicles make microscopic driving decisions are in consideration of the safety factors of their surrounding environment. In order to keep safe, when the leading vehicle decelerates, the following vehicle will usually decelerate or change lanes, if safety conditions are satisfied. In other words, the safety factors directly influence the microscopic behavior of vehicles, and the driving risk potential field can directly express this safety factor in the form of potential field strength. Therefore, the key vehicle motion information which was neglected in the traditional environment is taken into account in the modeling process of the driving risk potential field model and then we develop a dynamic driving risk potential field-based model to capture the safety factors of microscopic driving behaviours. With this basic consideration, we applied this dynamic driving risk potential field model to the car-following and lane-changing models. The results illustrate that the driving risk potential field-based model can successfully predict the rules of microscopic behavior and can be applied to the CAVs environment. It opens up a new way of thinking in the direction of driving risk potential field theory modeling under the CAVs environment and its application in the microscopic model.

The rest of the paper is organized as follows. In Section II, the concept of the driving risk potential field is introduced first, and then the characteristics of the driving risk potential field and its classification are described separately. Section III presents the modeling process of the proposed driving risk potential field under the CAVs environment in detail and expounds its dynamic characteristics while analyzing the model expression. In Section IV, a car-following model is constructed based on the driving risk potential field model. In Section V, real-world NGSIM data is selected and used to calibrate the parameters of the proposed car-following model. The model's effectiveness is verified by comparison with OVM and IDM. Section VI presents the application of the driving risk potential field in the safety criterion of lane-changing behavior. Section VII concludes this paper and discusses future work.

II. Concept of the Driving Risk Potential Field Model

A. Characteristics of the Driving Risk Potential Field

According to Physics theory, the potential field consists of two concepts: Field and Potential. First, the Field can be understood as a prediction of the ability of a certain object with defined attributes to interact with other objects around it. The prediction here means that if there are other objects near this certain object, there will be mutual actions between them. Otherwise, there will be no mutual action (if other objects don't exist). For example, if the defined attribute of an object is mass or a charge then the object has a gravitational field or an electrostatic field, respectively. It should be noted that the magnitude of the field value is related to the state of the object with defined attributes. Therefore, the Field of an object itself is only related to its own defining attributes and state, and it is a description of the mutual action of the whole space around the object. No matter whether there are other objects around, it will not affect the Field formed by it. The prediction of a point around an object is called the Potential, that is to say, the Potential can be used to predict the mutual action of other objects at one point. The magnitude of the mutual action refers to energy. The energy of an object due to its position in the Field is called potential energy. In conclusion, the Potential Field in physics has three basic characteristics [18]:

1) Objectivity: The potential field exists objectively in nature. The potential field can be classified according to different defining attributes of objects, such as gravitational field, electromagnetic field, temperature field, etc.

2) Universality: From the study of all kinds of potential fields, it is found that any kind of potential field exists universally. As long as the defining attribute of an object is not lost, the potential field determined by that attribute will exist universally.

3) Measurability: The magnitude of the potential field is only related to its own defining attributes and state. For example, in the gravitational field, the corresponding defining attribute is the mass of the object, and the state of the object corresponds to its motion state. Therefore, the magnitude of the gravitational field in different motion states can be quantitatively measured according to the mass of the object.

From the perspective of traffic flow theory, there exists a mapping between microscopic driving behavior and potential field theory. The microscopic driving behavior of the traffic flow mainly includes car-following and lane-changing behavior. Because the vehicle is affected by its surrounding environment in the process of driving, it will make different driving behavior decisions on the premise of maintaining its own safe driving. Generally speaking, no matter what kind of driving behavior a vehicle makes, its purpose is to achieve its desired velocity on the premise of ensuring a safe distance between itself and other vehicles. Taking single-lane car-following behavior as an example, when the following vehicle keeps a long distance from the leading vehicle and the following vehicle does not reach its desired velocity, this following vehicle will choose to accelerate until it reaches its desired velocity or stops accelerating when the distance between the two vehicles is reduced to a certain threshold before reaching the desired velocity. At this time, when the distance between the two vehicles continues to decrease, the following vehicle will choose to decelerate to prevent traffic safety accidents. Similarly, for lane-changing behavior, when an object vehicle performs lane-changing behavior, the main factor determining whether the lane-changing behavior can be successful is not the object vehicle itself, but the space position of other vehicles in the target lane and the motion state of those vehicles. According to Newton's first law, "Every object persists in its state of rest or uniform motion in a straight line unless it is compelled to change that state by external force impressed on it". Although the object vehicle does not have substantial physical contact with the surrounding vehicles in traffic microscopic behavior, its motion state is changed by the position and motion state of the surrounding vehicles, which indicates that there is a virtual "external force" acting on the object vehicle, forcing it to change its motion state. This virtual force provided by the surrounding vehicles belongs to the field force because it isn't generated by contact and is determined only by the space position. The above shows that there is a potential field in real traffic microscopic behavior and even in the traffic system, which represents the risk of vehicle driving safety [6].

According to the above description of potential field theory in traffic microscopic behavior, we can define the general physical potential field in traffic flow as a driving risk potential field which describes the driving risks faced by vehicles in the driving process, and it represents the impact for vehicle safety from every traffic subject. That is to say, every traffic subject (not limited to vehicles, it can be: pedestrians, obstacles, signal lights, etc.) in the traffic environment will form its own driving risk potential field. The value of every traffic subject's own driving risk potential field is determined by their own attributes, and the driving risk potential field generated by each traffic subject will produce a field force for the others. Which means the multiple field forces that each traffic object receives come from different driving risk potential fields beside itself, and the superposition of these field forces is the main factor that leads to the change of individual motion state. Therefore, the characteristics of the driving risk potential field in traffic are consistent with the basic characteristics of other physical potential fields. It also has three main characteristics: objectivity, universality, and measurability.

B. Classification of the Driving Risk Potential Field

The road traffic environment is composed of people, vehicles, roads, environment, and other traffic infrastructures. The total road driving risk potential field is the superposition of many traffic elements' driving risk potential fields. In a simple road environment, there are two kinds of road marking lines; one is the lane dividing line which is used to separate vehicles running in the same direction. The other is the double amber line that divides vehicles running in the same direction from vehicles running in the opposite direction. Generally, vehicles

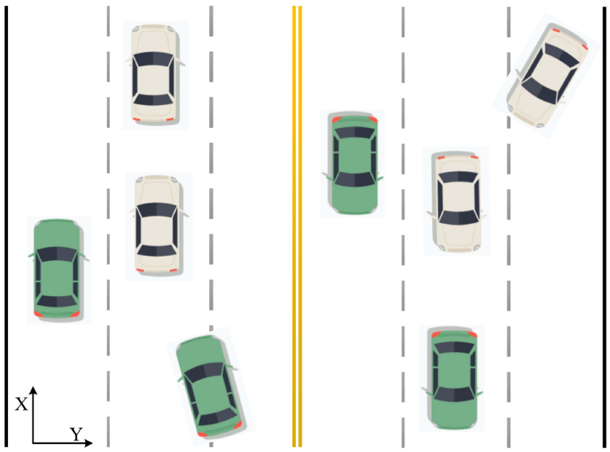


Fig. 1. Composition of road environment.

are not allowed to travel across this line. There are also exit road boundaries where vehicles are strictly prohibited. As such, the driving risk potential field in this simple road environment can be divided into three categories: the lane marking potential field, the road boundary potential field, and the vehicle potential field. Each potential field has its own field intensity function, which is: E_L, E_B and E_V. The road marking lines, road boundaries and vehicles in the actual road environment could be regarded as vector fields of multiple field sources. In the vector field, there are cases where the field strengths will cancel each other due to different vector field strength directions. However, this phenomenon does not exist in the safety risk of the traffic environment. The road safety risk after the superposition of different driving risk potential fields will only increase without decreasing, and the weight of different types of safety potential places in some special scenarios will be different. For example, when the vehicle is changing lanes, the weight of the lane marking potential field will be much smaller than that of the vehicle potential field. This is because the road marking line has a small impact on the object vehicle when it is allowed to change lanes, compared with other vehicles' potential fields in the target lane. Since we only concentrated on the study of car-following, the distribution of these weights in different situations will be the content of our subsequent research. Thus, the intensity of the driving risk potential field at a certain point on the road is defined as the weighted sum of the field strength modes generated by different field sources. Therefore, the total field intensity of driving risk potential field E_{total} can be expressed as:

$$|E_{\text{total}}| = \omega_L |E_L| + \omega_S |E_S| + \omega_V |E_V| \tag{1}$$

III. Modeling of the Driving Risk Potential Field

A. Lane Marking Potential Field

As previously mentioned, the road line potential field is comprised of the lane dividing line potential field and the double amber line potential field. Compared with the lane dividing line, the potential field generated by the double amber lines is much higher, because vehicles are generally prohibited from crossing this line. According to the above analysis, the field intensity function of the lane marking potential field can be expressed as:

$$E_L = \sum_{i,j=1}^{n} A_i e^{(-\frac{|d_{Aj}^L|^2}{2\sigma^2})} \cdot \frac{d_{Aj}^L}{|d_{Aj}^L|}$$
$$d_{Aj}^L = y_{l,j} - y_A \tag{2}$$

where A_i represents the field intensity coefficient of different types of lane marking, which determine the maximum peak value of lane marking potential field intensity. For example, the value of double amber lines A_2 is much larger than that of lane dividing line A_1. y_A is the ordinate value of point A in the road and $y_{(l,j)}$ represents the position coordinates of j_{th} lane marking along the Y-axis; d_{Aj}^l represents the distance vector from point A to j_{th} lane marking; σ determines the velocity at which the potential field increases or decreases as the vehicle approaches or moves away from the lane marking.

TABLE I
Parameter Values

Parameter	Values
A_1	2
A_2	8
σ	1.22
η	3

B. Road Boundary Potential Field

The road boundary potential field is generated in part of the road boundary line and extends to infinity with the vehicle approaching gradually. The purpose is to prevent the collision danger caused by the vehicle approaching. Therefore, the calculation of the road boundary field function can be expressed as:

$$E_B = \sum_{j=1}^{2} \frac{1}{2}\eta(\frac{1}{|d_{Aj}^B|})^2 \cdot \frac{d_{Aj}^B}{|d_{Aj}^B|}$$
$$d_{Aj}^B = (y_{B,j} - y_A) \tag{3}$$

where y_{left} and y_{right} are the position coordinates of the road boundary on the left and right sides, respectively. And η is the coefficient of the road boundary potential field. According to the research results of Wolf and Burdick [1], the values of those parameters of the road line potential field and the road boundary potential field are shown in Table I.

Based on Eq. (1) - (2), the lane marking potential field, the road boundary potential field and the superposition potential field are shown in Fig. 2. Fig. 2(a) displays the plane figure of the lane marking potential field (blue lines), the road boundary potential field (yellow lines) and the superposition potential field (green lines). For the lane marking potential field, we can find that the potential field intensity of the lane dividing line is much lower than that of the double amber lines, and they both achieved the maximum in their actual position of the Y-axis. For the road boundary potential field,

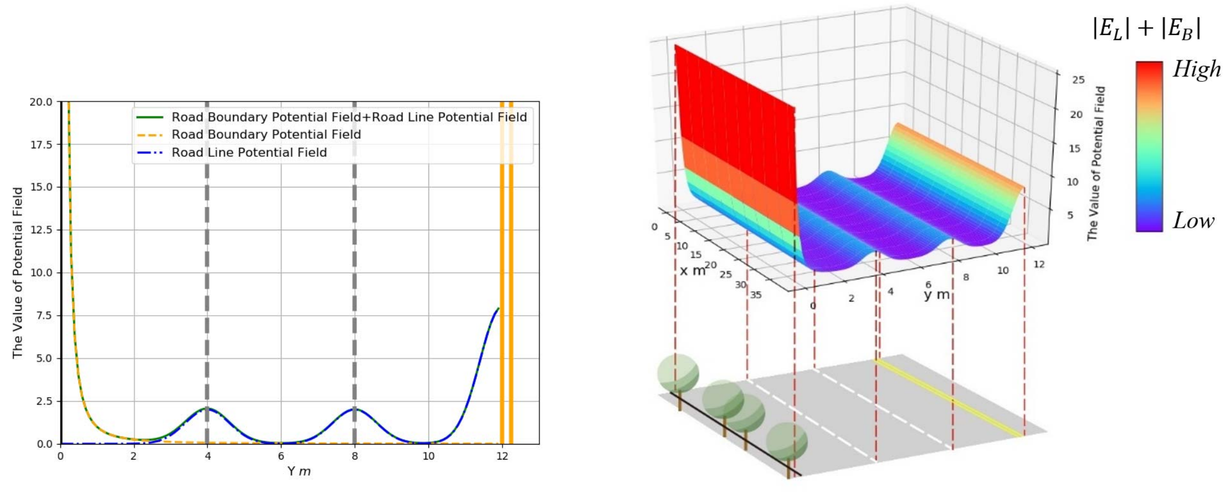


Fig. 2. Distribution map of road line field and road boundary field.

the potential field intensity of the road boundary decreases with the distance away from the road boundary, and it reaches infinity at the actual location of the Y-axis on the road boundary. Fig. 2(b) shows the superposition potential field in three dimensions. It should be noted that this superposition potential field represents only the road marking and road boundary potential field intensity without vehicles, obstacles and transportation facilities. It can be found that the potential field intensity is the lowest in the middle of each lane, that is, the vehicle is required to remain as much as possible in the middle of the road. Even if it is necessary to perform a lane-changing behavior, it must be carried out in areas with low potential field intensity. This is because the potential field intensity represents the magnitude of the driving risk factor, and the driving risk increases with the increase of potential field intensity.

C. Vehicle Potential Field

The vehicle potential field derives from the vehicle on the road. The intensity of the potential field formed by one object vehicle is determined by its defined attributes (e.g., vehicle type, vehicle mass) and motion state (e.g., velocity, acceleration). Details of these parameters are summarized in the following.

1) Vehicle Attributes: Generally, the mass of a vehicle is positively correlated with its type (e.g. in general, the mass of trucks is greater than that of SUVs (sport utility vehicles) and the mass of SUVs is usually greater than that of cars). Therefore, the impact of vehicle type on the vehicle potential field will not be considered in this paper. Through the study of [6], [7], it is found that the driving risks of vehicles are different under different velocities, even if they are of the same quality. The mass of a vehicle in motion can be defined as a kind of equivalent mass, which is related to the current velocity of the vehicle and reflects the driving risk degree of the vehicle at a certain speed. The mathematical expression of equivalent mass proposed in [6], [7] is adopted, which is defined as.

$$M_i = m_i \cdot (1.566 \times 10^{-14} \cdot v^{6.687} + 0.3345) \tag{4}$$

where M_i denotes the equivalent mass of object vehicle i, and m_i is the actual mass of object vehicle i. It can be seen that the equivalent mass increases with the increase of velocity, that is, the safety risk of the object vehicle at high speed is much greater than the vehicle at low speed.

2) Vehicle Motion State: A vehicle in the driving risk potential field of an object vehicle will experience different field intensity (or field force), which is related to its relative location with the object vehicle. The field intensity denotes the current safety risk degree of the vehicle imposed by the object vehicle. When the object vehicle is in different motion states, its driving risk potential field intensity will be different. Velocity v and acceleration a of the object vehicle are the main factors affecting its potential field intensity.

- Along the direction of its movement (vehicles in the same lane)

When the object vehicle moves at a constant speed, the safety risks received by the preceding vehicle (Vehicle A in Fig. 3) and the lag vehicle (Vehicle B in Fig. 3) should be consistent. However, when the object vehicle accelerates, the safety risk of the preceding vehicle is much greater than that of the rear vehicle, if they are still driving at the current speed. On the other hand, when the object vehicle decelerates, the safety risk of the lag vehicle is greater than that of the preceding vehicle. It should be noted that the safety risk will increase with the increase of velocity and acceleration on the object vehicle.

- Other directions (vehicles in other lanes)

When vehicles in other lanes do not change lanes (e.g., vehicle B and vehicle C in Fig. 3), they are less affected by the object vehicle than the preceding and lag vehicles in the same lane as the object vehicle. This is due to the lane dividing lines, as mentioned in Section III.

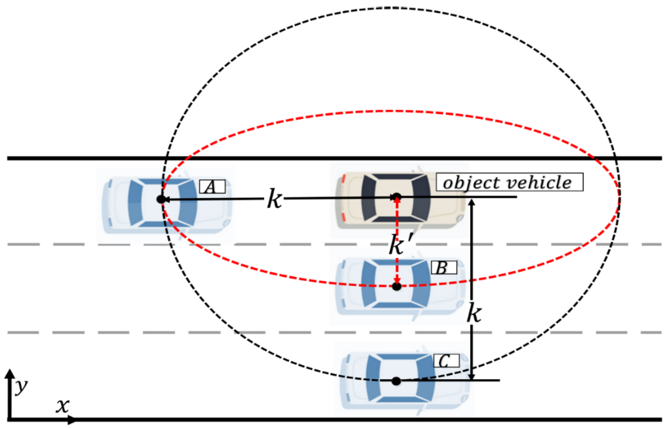


Fig. 3. Description of distance and pseudo-distance.

3) Distance Away From Vehicle: Generally, in the plane of the potential field, the field strength at any point is related to the distance from the object vehicle. The field intensity will increase as the distance decreases and reaches the maximum at the object vehicle position. Let the space coordinates of the object vehicle's centroid be (x_0, y_0). The distance k from any point (x,y) to the vehicle's centroid in space can be expressed as:

$$k = \sqrt{(\mathrm{x}-\mathrm{x}_0)^2+(\mathrm{y}-\mathrm{y}_0)^2} \tag{5}$$

If we use 1/k to directly represent the negative correlation between distance and field strength it can be found that the field intensity is the same at equidistant points, no matter whether the vehicle is in the same lane as the target vehicle or in another lane (e.g., vehicle A and D in Fig. 3). However, this is inconsistent with reality. In practice, a vehicle should stay far behind a leading vehicle in its lane, but is allowed relatively near the obstacle to the side or at the road boundary, that is, the degree of safety risk is similar when the vehicle is 30 meters behind the object vehicle and 2 meters to the side of the object vehicle [1]. The reason for this is that there is no velocity component in the vertical direction to the driving direction of the vehicle. Therefore, in order to better describe the relationship between the safety field intensity and the distance, the concept of pseudo-distance is introduced in this paper, and the distance in real space is altered to better describe the change of safety risk degree when a vehicle approaches the object vehicle from different angles. In the space coordinate system, we assume that the vehicle is moving along the x-axis, the vertical direction, while the driving direction of the vehicle is the y-axis.

Based on the analysis above, the function of pseudo-distance k' can be expressed as:

$$k' = \sqrt{\left[(x - x_0)\,\frac{\tau}{e^{\alpha v}}\right]^2 + [(y - y_0)\,\tau]^2} \tag{6}$$

where τ is the critical threshold of the safe distance and α is the undetermined parameter related to velocity.

In the driving process, the range of influence is limited for an object vehicle, that is, it will only affect several vehicles around it. The influence will decrease rapidly with increasing distance. In physics, we can regard it as a short-range force. The Yukawa potential is proposed by Japanese physicist Yukawa Hideki to describe the short-range interaction between particles.

The interaction between two vehicles is similar to that of two interacting particles described by the Yukawa potential. Motivated by this and the influence factor analysis above, we proposed a vehicle potential field function including the defining attributes and the motion state information, which can be expressed as:

$$\begin{aligned} E_v &= M_i \lambda \frac{e^{-\beta_1 a \cos\theta}}{|k'|} \cdot \frac{k'}{|k'|} \\ &= m_i \times \left(1.566 \times 10^{-14} v^{6.687} + 0.3345\right) \lambda \\ &\quad \times \frac{e^{-\beta_1 a \cos\theta}}{\sqrt{\left[(x - x_0)\frac{\tau}{e^{\alpha v}}\right]^2 + [(y - y_0)\tau]^2}} \cdot \frac{k'}{|k'|} \end{aligned} \tag{7}$$

where λ and β_1 are undetermined coefficients, θ is the clockwise angle formed by any point around the object vehicle and the mass center of the object vehicle with the motion direction of the vehicle and a is the acceleration of the current motion state of the object vehicle.

D. Characteristics Analysis of the Driving Risk Potential Field

Based on the analysis above, several moving vehicles and the field intensity of their corresponding potential fields are shown in Fig. 4. We can find that vehicles have different forms of potential fields due to their different motion states (i.e. different velocities and accelerations). In order to describe the distribution of vehicle potential fields under different motion states, the isopotential surface projection of the vehicle potential field of nine different motion states is listed in detail, as shown in Fig. 5.

In Fig. 5, each black contour represents a horizontal curve formed by projecting points with the same potential field into a plane, ϕ refers to the angle between the direction of the vehicle velocity and the X-axis (counter-clockwise), which can also be understood as the steering angle of the vehicle. The depth of the color indicates the intensity of the potential field. Although different motion states exhibit different potential field distributions, some of the common characteristics and changing laws cannot be changed in any motion state.

1) The intensity of the potential field reaches the maximum at the location of the object vehicle and approaching to infinity, which means that if other vehicles appear at this location, collisions will inevitably occur and eventually lead to accidents.

2) As the relative distance between vehicles increases, the potential field intensity will gradually decrease. When the relative distance exceeds a certain threshold, the intensity of the potential field is extremely small and can be ignored. This means that even if other vehicles appear in these areas, these vehicles will not face any driving risks from the object vehicle. This result is completely consistent with the analysis of the

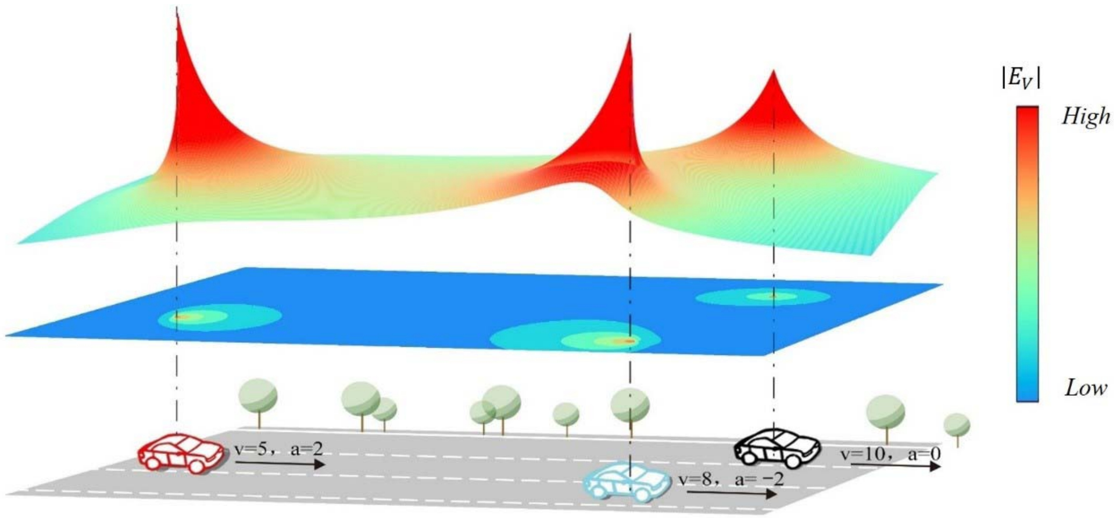


Fig. 4. Corresponding distribution of vehicle potential fields in a traffic scenario.

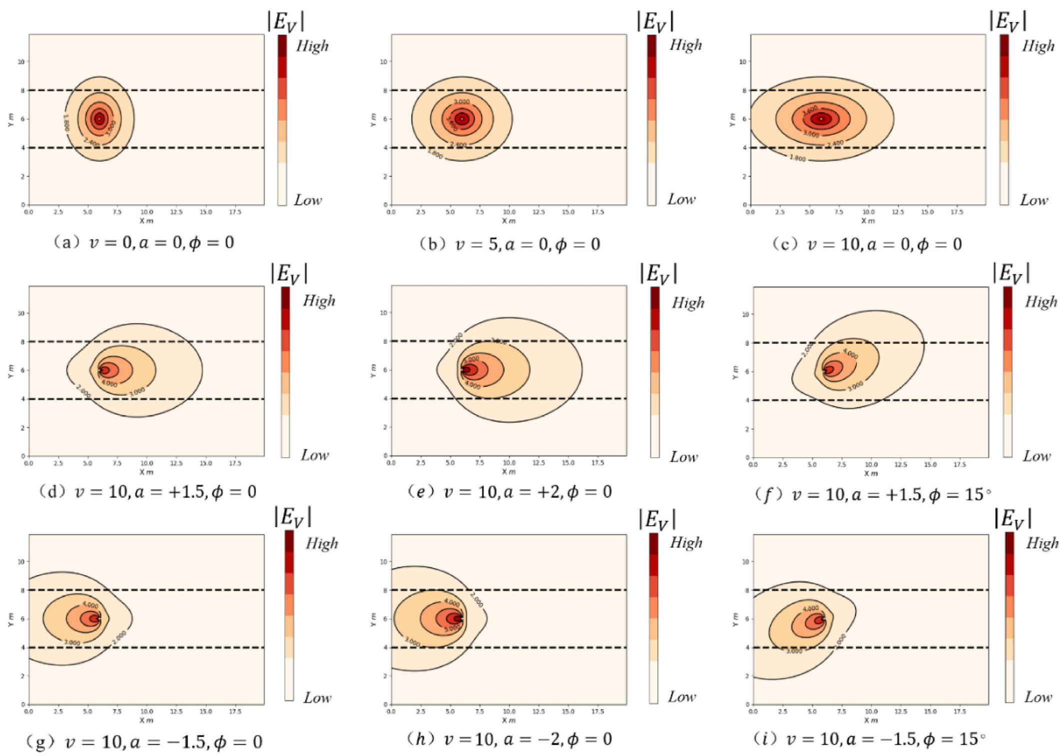


Fig. 5. Isopotential surface projection of the driving risk potential field.

influence factors of the vehicle potential field and the analysis of the "short-range force" mentioned above.

In addition to these common characteristics, there is no doubt that they have other different characteristics, which are analyzed as follows.

1) When the object vehicle is in a stationary state, as shown in Fig. 5(a), the isopotential surface projection of the vehicle potential field in this scenario is composed of several concentric circles, which means that the safety risk level is the same at the same distance regardless of the angle of the other vehicle approaching the object vehicle. In this case, the vehicle can be regarded as an obstacle.

2) In Fig. 5(b)-(c), the vehicle moves at a constant speed along the x-axis. Compared to the first scenario in Fig. 5(a), the vehicle's potential field intensity in the Y-axis direction remains the same value regardless of the velocity change in the X-axis direction. On the contrary, the vehicle potential field intensity increases with the increase of the object vehicle's velocity in the X-axis direction. It can be clearly found that as the speed of the object vehicle increases, the potential

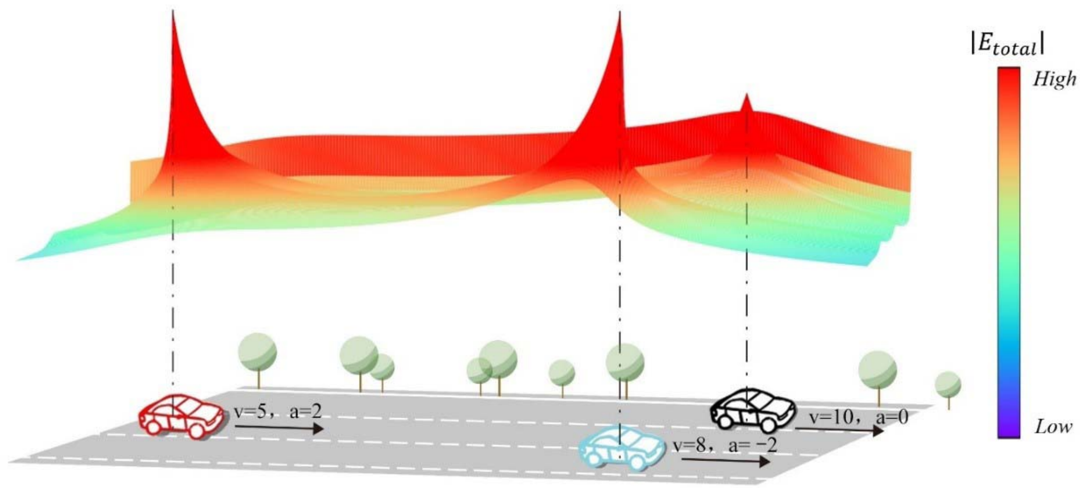


Fig. 6. Corresponding distribution of vehicle potential fields in a traffic scenario.

field distribution gradually expands along its direction of movement. Therefore, in the scenario of uniform velocity, the isopotential surface projection of the vehicle potential field is composed of several concentric ellipses, as shown in Fig. 5(b)-(c);

3) Fig. 5(d)-(e) describe the acceleration behavior of the vehicle along the x-axis. The vehicle's velocity in both of these scenarios is 10 m?s. It can be found that the distribution of the vehicle potential field is obviously different from that of the vehicle under the uniform state. The distribution of the vehicle potential field in these two scenarios is obviously inclined to the front, that is, the front area of the object vehicle in its direction of movement will become the main area affected by the vehicle's movement, which indicates that the impact of the object vehicle on the preceding vehicle will be much greater than the impact on the lag vehicle when it accelerates. This "forward incline" feature will become more pronounced as the acceleration increases. It is completely consistent with the actual driving situation. In the future CAVs environment where the motion state information of vehicles can be shared in real-time, if the object vehicle is in an accelerated state and there is a critical safety distance to the following vehicle, then the following vehicle can keep at a constant velocity or even accelerate within the allowable safety ensuring range depending on the object vehicle's potential field value at its position. Therefore, timely decisions of the following vehicle based on the information of the object vehicle (leading vehicle) will improve the road capacity, to some extent.

4) Fig. 5(g)-(h) describe the deceleration behavior of the vehicle. The velocity in both of these scenarios is 10 m?s. The distribution of the vehicle potential field in these two scenarios exhibits a "backward incline" characteristic, which is completely opposite to the "forward incline" characteristic shown in Fig. 5(d) and (e). This is also exactly consistent with the actual driving situation. Similarly, in the CAVs environment, if the object vehicle is in a deceleration state, the following vehicle can decelerate in advance according to the object vehicle's potential field value at its position, even if there is a large safety distance to the object vehicle. Undoubtedly, this will bring some improvement to road safety.

5) Fig. 5(f)-(i) represent the scenarios when the object vehicle changes its direction of movement under different acceleration values (e.g., preparing for a lane change). It could be found that the distribution of the potential field will change at the same angle as the vehicle's direction of movement changes. Other characteristics are also determined by the speed and acceleration of the object vehicle, as described above.

According to Eq. (1), the field intensity function of total driving risk potential field E_{total} in complex traffic, the scenario is composed of three parts: lane marking potential field intensity E_L, road boundary potential field intensity E_B and vehicle potential field intensity E_V. Based on the expression of these potential fields above, a complex traffic scenario and the total driving risk potential field intensity are shown in Fig. 6. The impact of traffic factors on driving safety can be assessed visually by the distribution of the field intensity, which is represented by different colors in Fig. 6. The darker color indicates that the road environment is in a more dangerous situation.

IV. Application in the Car-Following Model

The microscopic models describe traffic flow dynamics in terms of single vehicles. Microscopic traffic flow modeling has always been one of the most important issues in traffic flow theory. The vehicle dynamics are mainly embodied in two microscopic behaviors: car-following and lane-changing. Through the above analysis, it can be concluded that the driving risk potential field model can dynamically describe the motion state of each traffic subject in a complex traffic environment. The object vehicle can determine its microscopic motion behavior according to the distribution of the total potential field of its surrounding traffic environment. This section will introduce the application of the driving risk potential field theory on the car-following model in detail.

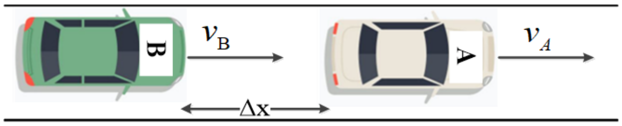


Fig. 7. Illustration of the single-lane car-following scenario.

A. Traditional Car-Following Models

The research of the car-following model is of great significance to understand the characteristics of traffic flow. The analysis results of the car-following model can be applied to traffic planning, traffic management, and traffic control, and then improve the existing traffic facilities environment, and solve many practical problems of the existing road traffic. From the perspective of traffic engineering, car-following models can be divided into four groups [19]: The stimulus-response group: e.g. the GM model [20], the Newell model [21], and the Helly model [22]. The secure distance group: e.g. the Gipps Model [23] and the FRESIM Model [24]. The psycho-physiology group: e.g. the Wiedemann model [25], the Winsum model [26], and the DVA model [27]. And the artificial intelligence group: e.g. neural network models [28], [29]. While from the point of view of statistical physics, car-following models can be divided into three kinds of models: The optimization velocity models, such as the OVM (Optimal Velocity Model) [30], the GF model [31], and the FVD model [32]. The intelligent driving models, such as the IDM (Intelligent Driver Model) [33]. And the cellular automata models, such as the NaSch model [34], and the FI model [35]. Those car-following models have attracted the attention of many scholars in recent years, who conducted their research from different viewpoints. Aghabayk *et al.* optimized the car-following model considering the influence of different vehicle types on car-following behavior [36]. He *et al.* and Papathanasopoulou *et al.* made innovations in research methods based on traffic big data and constructed different data-driven car-following models [37], [38]. Lu *et al.* considered the impact of individual driving behavior on car-following behavior and proposed a car-following model based on the quantification of the driving risk faced by drivers [39].

B. Establishment of the Car-Following Model Based on the Driving Risk Potential Field

Fig.7 shows a typical car-following scenario; there are two vehicles, leading vehicle A and following vehicle B, and Δx represents the distance between the leading and following vehicles. Assume that both the leading and following vehicles are traveling along the middle line of the road, thus the road line potential field and the road boundary potential field can be ignored. As such, the driving risk potential field affecting vehicle B only comes from the vehicle potential field of vehicle A. In this scenario, the vehicle potential field intensity of vehicle A can be obtained by simplifying Eq. (7) without considering the factors of lane change (only consider the direction of motion of the vehicle) and angle θ, the vehicle potential field intensity of vehicle A can thus be expressed as:

$$\begin{aligned} E_v^A &= M_A \lambda \frac{e^{-\beta_1 a_A}}{|k'_x|} \cdot \frac{k'_x}{|k'_x|} \\ &= M_A \lambda \frac{e^{-\beta_1 a_A}}{\sqrt{\left[(x-x_0)\frac{\tau}{e^{\alpha v_A}}\right]^2}} \cdot \frac{k'_x}{|k'_x|} \\ &= m_A \times \left(1.566 \times 10^{-14} {v_A}^{6.687} + 0.3345\right) \\ &\quad \times \lambda \frac{e^{-\beta_1 a_A + \alpha v_A}}{\tau \cdot \Delta x} \cdot \frac{k'_x}{|k'_x|} \end{aligned} \tag{8}$$

The field force F of a vehicle in the real car-following state represents the degree of safety risk that the vehicle receives, and the higher the safety risk, the greater the field force that the vehicle receives. The field force is mainly determined by the potential field intensity, the vehicle motion state, and the vehicle attributes. These factors are summarized as follows:

(1) Potential field intensity: Force F is a repulsive force; its purpose is to prevent vehicles from approaching other objects, including non-moving obstacles and moving vehicles. Generally, a greater potential field intensity indicates a higher safety risk, thus the field force F received by the vehicle is positively correlated with the potential field strength.

(2) Vehicle motion state: The vehicle motion state is mainly determined by its velocity. When the velocity is higher and the direction of the velocity matches the direction of the potential field intensity (when the vehicle is constantly close to the obstacle or surrounding vehicles), the field force F will be greater; when the direction of the velocity is opposite to the potential field intensity (when the vehicle is away from the obstacle or the surrounding vehicles), the field force F will gradually decrease; When the vehicle is stationary, this factor will have no effect on the field force F.

(3) Vehicle Attributes: This refers to the actual mass of the vehicle. The actual mass of the vehicle can generally be used as a criterion for the classification of vehicle types (e.g. the actual mass of the truck is usually greater than that of the car). Under the same velocity conditions, the field force F of the vehicle with large mass is greater than that of the vehicle with small mass.

Through the above analysis, we can refer to the formula of electric field force: $F = Eq$, in which the field intensity E in the formula can be understood as the vehicle potential field intensity E_v^A; q is the quantity of electric charge, which belongs to the defined attribute in the electric field environment. In this paper, the defining attributes in the vehicle potential field are defined as the vehicle motion state (vehicle speed) and vehicle attributes (vehicle actual mass). Then the field force F_{AB} formed by the vehicle potential field strength E_v^A of the leading vehicle A to the following vehicle B can be calculated by Eq. (9), in which β_2 is the undetermined coefficient.

$$\begin{aligned} F_{AB} &= m_B \cdot e^{\beta_2 v_B} \cdot \left|E_v^A\right| \\ &= m_B \cdot e^{\beta_2 v_B} \cdot M_A \lambda \frac{e^{-\beta_1 a_A + \alpha v_A}}{\tau \Delta x} \end{aligned} \tag{9}$$

because the field force F_{AB} is the cause of changing the motion state of vehicle B. Therefore, the acceleration a_{AB} generated by the vehicle B under the action of the field force F_{AB} is as shown in Eq. (10).

$$a_{AB} = \frac{-F_{AB}}{m_B} = -M_A \lambda \frac{e^{-\beta_1 a_A + \alpha v_A}}{\tau \cdot \Delta x} \cdot e^{\beta_2 v_B} \tag{10}$$

According to the previous description of the driving risk potential field, the field force formed by the driving risk potential field belongs to the short-range force. This means the force will only work on the vehicle under a certain distance, so when the vehicles are far apart from each other, the vehicle's motion behavior is mainly affected by its desired velocity. If the desired velocity is not reached, the vehicle will accelerate. Otherwise, the vehicle will decelerate in the case where its velocity exceeds the desired velocity. Therefore, a velocity potential field can be defined to accurately describe such driving behavior. The velocity potential field function is mainly affected by the relationship between the desired velocity and the current velocity, it has the following expression:

$$E_{velocity} = |E_{max}|\, tanh\left[\delta \cdot \left(v_0^{(\alpha)} - v_B\right)\right] \tag{11}$$

In Eq. (11), $E_{velocity}$ is the potential field strength of the velocity potential field, $|E_{max}|$ represents the maximum potential field strength value of the velocity potential field, δ is the undetermined coefficient. $v_0^{(\alpha)}$ is the desired velocity of vehicle B, v_B represents the velocity of vehicle B at the current moment. Then the field force $F_{velocity}$ formed by the velocity potential field strength $E_{velocity}$ can be calculated by Eq. (12), in which $|F_{max}|$ is the maximum field force value caused by the maximum potential field strength E_{max}.

$$F_{velocity} = |F_{max}|\, tanh\left[\delta \cdot \left(v_0^{(\alpha)} - v_B\right)\right] m_B \tag{12}$$

Thus the acceleration of the vehicle is mainly described by the relationship between the current velocity and the desired velocity of the vehicle. In this paper, we consider using Eq. (13) to describe this relationship.

$$a_{free} = a_{max} tanh\left[\delta \cdot \left(v_0^{(\alpha)} - v_B\right)\right] \tag{13}$$

In Eq. (13), a_{max} represents the maximum allowable acceleration of the vehicle, a_{free} represents the acceleration of vehicle B when the distance between vehicle B and the leading vehicle of vehicle B is long enough. At this time, the acceleration of vehicle B is only related to $v_0^{(\alpha)}$ and v_B. Eq. (13) depicts that when $v_B < v_0^{(\alpha)}$, the acceleration is positive, in the next moment vehicle B is in an accelerated state; when $v_B > v_0^{(\alpha)}$, the acceleration is negative, in the next moment vehicle B is in a decelerated state. Finally, the car-following model based on the theory of the driving risk potential field (In order to facilitate the description, the model is abbreviated as the driving risk potential field model (DRPFM)) in this paper is given by Eq. (10) and (13):

$$\begin{aligned} \dot{v_B} &= a_{free} + a_{AB} \\ &= a_{max} \tanh\left[\delta \cdot \left(v_0^{(\alpha)} - v_B\right)\right] \\ &\quad - M_A \lambda \frac{e^{-\beta_1 a_A + \alpha v_A}}{\tau \cdot \Delta x} \cdot e^{\beta_2 v_B} \end{aligned} \tag{14}$$

In the appropriate car-following model, the acceleration functions encoding the driving behavior should at least model the following three basic requirements [40]: The acceleration is a strictly decreasing function of the velocity; the acceleration is an increasing function of the distance to the leading vehicle; the acceleration is a decreasing function with the velocity of approach to the lead vehicle. The DRPFM established in this paper satisfies the above requirements, and the proof is provided in Appendix.

V. Model Calibration and Evaluation

The parameter calibration process of the model refers to the process of concretizing the undetermined parameters in the model through effective mathematical methods combined with actual vehicle data. Many parameters have been defined in our proposed car-following model, whose values are of significant importance for our evaluation of driving safety risk. Thus, effective parameter calibration needs to be conducted in this section to enhance the applicability of the model and avoid model distortion.

A. Data Description

Under the CAVs environment, vehicles' motion information can be transmitted and shared in real-time. And vehicles can make driving decisions based on such numerical information. In the traditional environment (human-driven vehicle), although the driver cannot get the specific values of these motion information, they will be very sensitive to them especially in some special scenarios, e.g., the stop (sudden deceleration) and go (sudden acceleration) process of the lead vehicle. In these scenarios, the acceleration value of the lead vehicle will change obviously. Although the following vehicle cannot get the specific value of these accelerations, the driver will make corresponding driving decisions according to his/her own sensitivity to these accelerations. Therefore, these acceleration-sensitive scenarios in the traditional environment could be regarded as scenarios similar to the CAVs environment, due to the lack of field data in real CAVs environments the actual data in the traditional environment can be filtered to obtain the vehicle trajectory data of these scenarios, which can be used to conduct the car-following model calibration. It should be noted that the parameters in the potential field model can also be determined by the calibration of the car-following model because the potential field model is the basis of our proposed car-following model.

NGSIM dataset is a reliable pure trajectory dataset, which is adopted by many scholars for micro-driving behavior studies. According to the purpose of this study, we select the trajectory data on the northbound direction of Interstate 80 (I-80) freeway in the San Francisco Bay area, California and the data record frequency is 0.1 s per frame.

B. Data Pre-Processing

Since the original NGSIM dataset contains lane change behavior data and also some non-car-following state data,

Leading-Vehicle

Vehicle_ID	Global_Time	Lane_ID	Local_Y,v_Vel...	Following
1	11188478718	2	167.928,19.99	3
1	11188478719	2	169.927,19.98	3
2	11188478720	3	215.167,22.33	4

Following-Vehicle

Vehicle_ID	Global_Time	Lane_ID	Local_Y,v_Vel...	Following
3	11188478718	2	81.973,20.0	0
3	11188478719	2	83.973,20.0	0
4	11188478720	3	126.029,21.05	0

Car-following pairs after splicing

Vehicle_ID	Following_ID	Global_Time	Lane_ID	Local_Y,v_Vel··· (Leading-Vehicle)	Local_Y,v_Vel··· (Following-Vehicle)
1	3	11188478718	2	167.928,19.99	81.973,20.0
1	3	11188478719	2	169.927,19.98	83.973,20.0
2	4	11188478720	3	215.167,22.33	126.029,21.05

Fig. 8. Data splicing process to obtain accurate car-following pair information.

it is necessary to perform some pre-processing work on the original dataset according to our research objectives before our calibration process.

At the outset, in order to study the car-following behavior, pairs of car-following information can be obtained through simple data connection processing due to the leading-vehicle ID and following-vehicle ID provided in NGSIM dataset, which is called "data splicing" process. It should be noted that we need to ensure that each car-following behavior information is in the same lane and at the same time according to the time-space information (Global_Time and Lane_ID) in the dataset. Fig. 8 represents the pre-processing of data splicing, each row of the data after processing represents the information of a car-following pair on a certain lane at a certain time.

After acquiring the data of car-following pairs, these data need to be screened to select car-following pairs with obvious acceleration changes, in order to better match the car-following behavior under CAVs environment as explained above. Finally, 412 sets of trajectory data of car-following pairs are obtained, containing the state of deceleration, parking, and start-up acceleration. Some of the trajectory data of car-following pairs are plotted as shown in Fig.9. It could be found that the following behavior of these car-following pairs aligns well with the reality, and there are no unreasonable phenomena such as overtaking and reversing. We can also find that a lot of stop-and-go driving process (the red dotted circle in Fig.9) are included in these car-following behaviors, which is completely consistent with our expected data pre-processing results and our research goals.

C. Selection of Calibration Algorithms

Artificial Bee Colony Algorithm (ABC) is one of the most recently defined algorithms by Dervis Karaboga for solving the problem of multivariate function optimization, motivated by the intelligent behavior of honey bees. It is adopted in this paper to solve our model calibration problem due to its fast convergence rate.

In the ABC algorithm, every feasible solution to the problem is called food sources, and the food picking process is the optimization process. In the algorithm, bees can be divided into three categories: the employed bees, the observe bees and the scout bees. There are four main phases in the algorithm: the Initialization Phase, the Employed Bees Phase, the Onlooker Bees Phase, and the Scout Bees Phase. The details of the ABC algorithms are presented in Algorithm 1 [41].

Algorithm 1 Artificial Bee Colony Algorithm

```
Initialization Phase
Cycle=1
repeat
        Employed Bees
        Onlooker Bees
        Scout Bees Phase
        Cycle = Cycle + 1
Until Cycle=Maximum
Cycle Number or Maximum CPU time
```

D. Calibration Results

The calibration of the car-following model can be regarded as a numerical optimization problem. Its goal is to minimize the gap between real vehicle trajectory data and predicted vehicle trajectory data through our proposed car-following model. The Root Mean Squared Error (RMSE) and the Mean Absolute Percent Error (MAPE) between real and predicted position of the lag vehicle is adopted to evaluate the effectiveness of our proposed model, which can be expressed as:

$$RMSE = \sqrt{\frac{\sum_{n=1}^{N} \left| x_n^{real} - x_n^{pre} \right|^2}{N}} \tag{15}$$

where x_n^{real} and x_n^{pre} represent the $n-th$ real and predicted position respectively, and N is the sample size calibrated for the model.

In order to better verify the effectiveness of our proposed model (DRPFM), the same data and the same ABC algorithm are applied to calibrate the parameters of OVM, IDM and DRPFM proposed in this paper. The calibration results are shown in Table II -Table IV.

E. Trajectory Estimation of DRPFM

Fig. 10-12 illustrate the comparison of the actual real trajectory with our simulation trajectory that is derived from

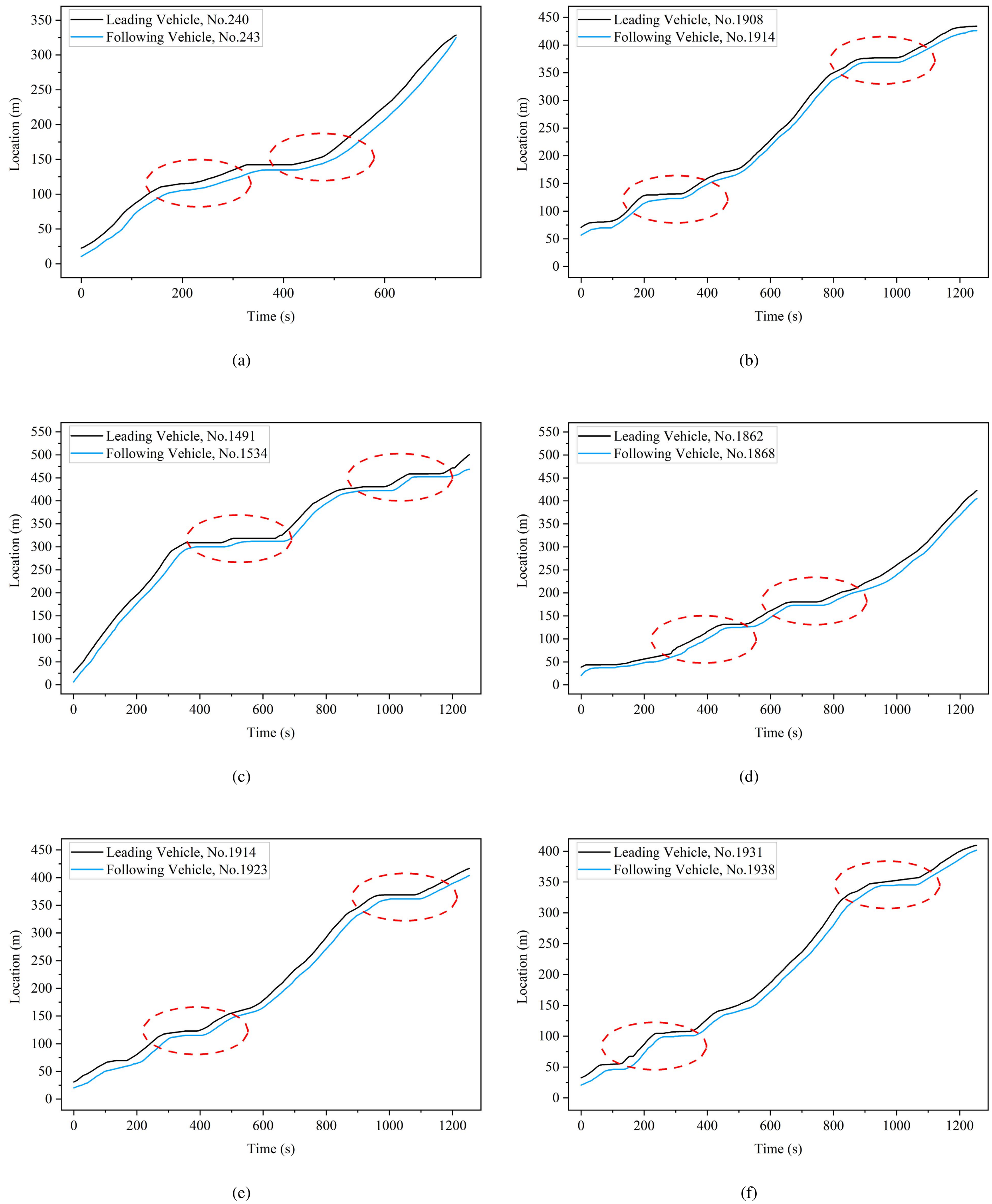


Fig. 9. Trajectory data examples of car-following pairs.

car-following models, the headway distance and vehicle velocity in the case of car-following scenarios. In the left part of the figure, the black and blue solid lines represent the real trajectory of the leading vehicle and the following vehicle, respectively. The red dashed line is the estimation trajectory of the following vehicle. It can be intuitively found that our proposed model and the IDM have good performance in describing the car-following behavior, which is significantly better than the OVM. In the right part of the figure, the red and blue lines indicate the headway distance and velocity of rear vehicle in real-world and the estimation that derived from car-following models, respectively. We can also find that our

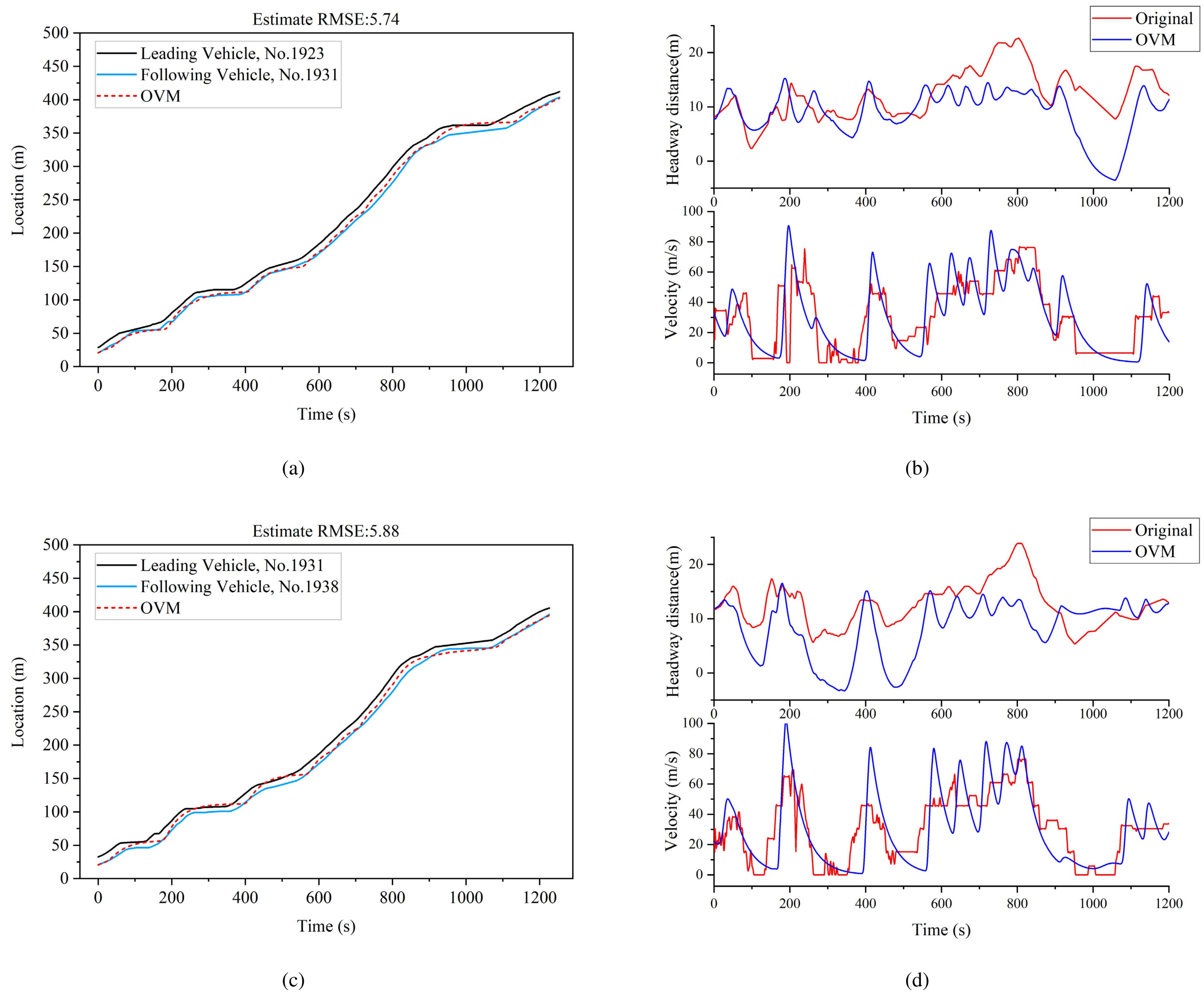


Fig. 10. The estimation performance of trajectory, headway distance and velocity of rear vehicle by OVM.

TABLE II
PARAMETER VALUE OF IDM

Parameter	Values
α_{ovm}	0.341
v_{max}^{ovm}	22.312
h_c	13.501

TABLE III
PARAMETER VALUE OF OVM

Parameter	Values
v_{free}	27.321
a	5.042
b	2.453
s_0	1.242
T	1.421

TABLE IV
PARAMETER VALUE OF DRPFM

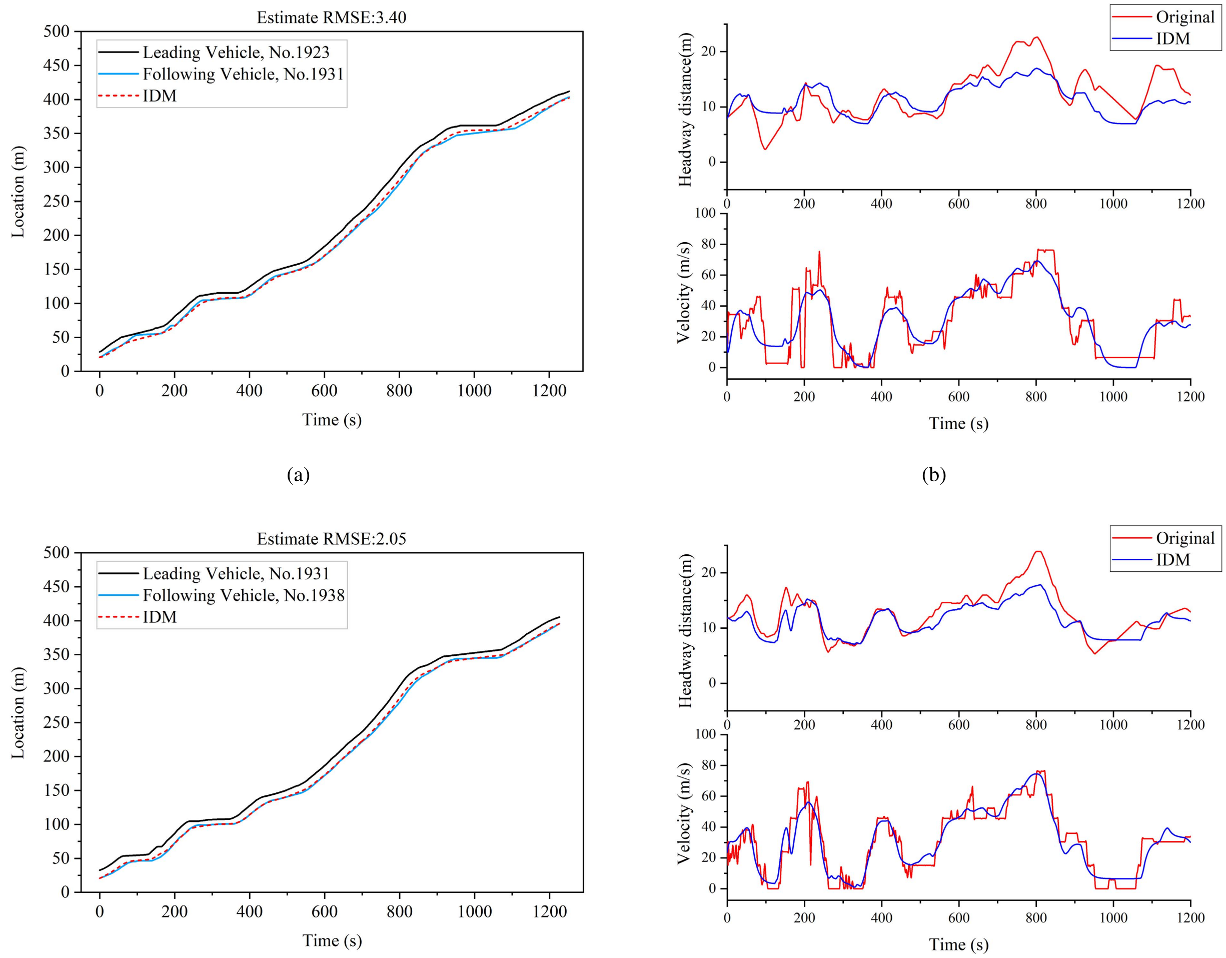
Parameter	Values
δ	4.647
$v_0^{(\alpha)}$	22.032
a_{max}	4.001
λ	0.056
α	0.029
β_1	-0.169
β_2	0.117
τ	2.717

proposed model and IDM are greatly superior to OVM. The performance of our model is generally consistent with that of IDM, but when we compare (c) and (d) in Fig.11 and Fig.12, it can be easily found that our model performs better at frequent oscillation phases, benefiting from our consideration of acceleration in the modeling process.

To compare the estimation performance of different car-following models more intuitively, the trajectory estimation performances are incorporated in the same graph, as shown in Fig.13. It can be seen from the error evaluation results that our proposed model DRPFM shows good performance.

In order to better compare the performance of the three models, in addition to RMSE, MAPE is also elected as the evaluation index for more effective evaluation, which can be expressed as:

$$MAPE = \left(\frac{1}{N} \sum_{n=1}^{N} \frac{\left| x_n^{real} - x_n^{pre} \right|}{x_n^{pre}} \right) \tag{16}$$



(a) (b)

(c) (d)

Fig. 11. The estimation performance of trajectory, headway distance and velocity of rear vehicle by IDM.

where x_n^{real} and x_m^{pre} represent the $m-th$ real and predicted position respectively, and N is the sample size calibrated for the model.

The values of RMSE and MAPE on every follower's trajectory of each model are shown in Fig. 13. As we can see, the DRPFM has the most accurate predictions compared with the OVM and the IDM in describing car-following behavior.

Among the three models, the error of the OVM is the largest; the error of the IDM and the DRPFM proposed in this paper are controlled within a small range, but the latter shows better precision than the IDM. In the past, in car-following models, the influence of acceleration on the car-following effect was rarely considered. This is because the acceleration parameters are difficult to acquire in real-time and shared among vehicles in the traditional traffic environment. But it is worth noting that while the lag vehicle is not able to feel the specific value of the acceleration of the preceding vehicle while performing the following behavior, it is indeed affected by the acceleration of the preceding vehicle, especially during the process of deceleration and the startup acceleration of the preceding vehicle. Since all the trajectory data selected in this paper have obvious phenomena of deceleration, stop and startup acceleration, the DRPFM proposed in this paper, which takes acceleration into account, therefore has the best accuracy compared with the OVM and the IDM. Most importantly, compared to the IDM, our proposed DRPFM performs better at the stage in which the vehicle movement state fluctuates greatly. Therefore, the DRPFM can be used as an alternative car-following model under the automated and connected environment in the future.

VI. Application of DRPFM in the Safety Criterion of Lane-Changing Behavior

A. The Safety Criterion of Lane-Changing Behavior

Fig. 14 depicts the general scenario of lane-changing. The subject vehicle (SV) is located in the center of the original lane; two alternatives are available for SV to select: keep following in the original lane or change to the target lane. Here and in the following, we denote the leading vehicle in the original lane as LV and the following vehicle in the original lane as FV. And we denote the gap of the following vehicle and the object vehicle as S_f and S_{SV}. In order to distinguish the old situation before the lane change and the

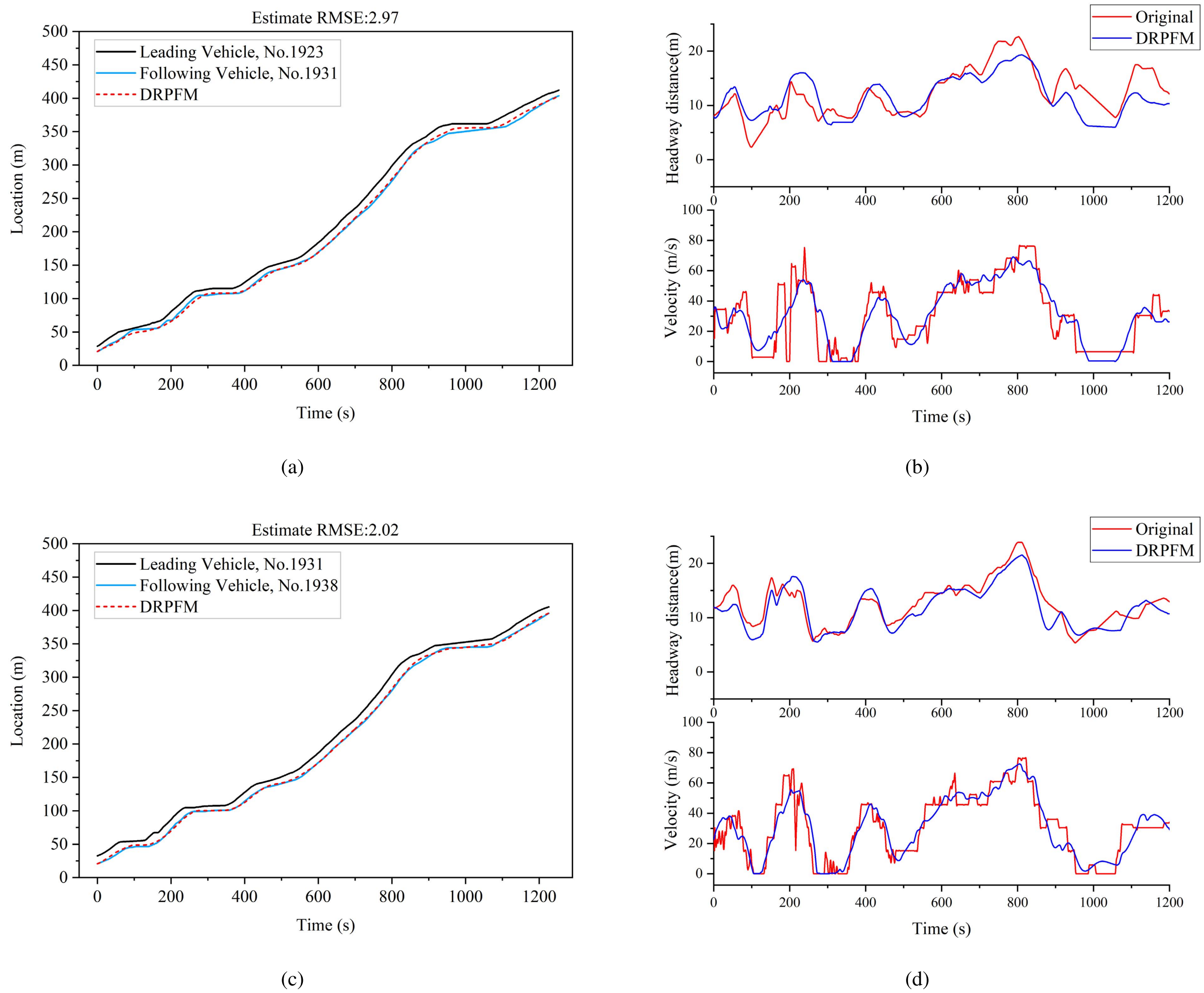


Fig. 12. The estimation performance of trajectory, headway distance and velocity of rear vehicle by DRPFM.

new situation after the lane change, a hat is added to those above symbols. For example,$\hat{L}V$ denotes the leading vehicle of SV in the target lane,$\hat{s}_f$ denotes the gap of the following vehicle in the new situation and $S_{\hat{f}}$ denotes the gap of the following vehicle in the old situation, $\hat{a}_{\hat{f}}$ denotes the acceleration of the following vehicle in the target lane in the new situation.

The safety criterion of the lane change is defined as Eq. (17) [40]. None of the drivers γ affected by the consequences of opting for lane-changing behavior should be forced to perform a critical maneuver as a consequence of a decision for lane-changing behavior. A maneuver is deemed to be critical if it entails braking decelerations exceeding the safe deceleration b_{safe}:

$$a^{\gamma} > -b_{safe} \tag{17}$$

Assume that the old situation (when the SV has not started the lane-changing process) is safe. Refer to the safety criterion Eq. (17), after the successful lane-changing process of the SV, the safety criterion of the $\hat{F}V$ in the new situation in the target lane becomes:

$$\hat{a}_{\hat{f}} = a\left(\hat{s}_{\hat{f}}, v_{\hat{f}}, v_{sv}\right) > -b_{safe} \tag{18}$$

B. Lane-Changing Rules Based on Car-Following Models

The lane-changing safety criterion mentioned above can combine with the longitudinal car-following models. In principle, the safety criterion is compatible with any car-following models which provide the acceleration function (For example the OVM and the IDM). The following sections will introduce the compatibility rules of safety criterion with the OVM and IDM, and then similar compatibility rules for the DRPFM are deduced.

1) Rules for OVM

The expression of the OVM is shown in Eq. (19) - (20).

$$a^{OVM} = \alpha_{ovm}\left[V\left(\Delta x_{i,i-1}(t)\right) - v_i(t)\right] \tag{19}$$

$$V\left(\Delta x_{i,i-1}(t)\right) = \frac{v_{max}^{OVM}}{2\left[\tanh\left(\Delta x_{i,i-1}(t) - h_c\right) + \tanh\left(h_c\right)\right]} \tag{20}$$

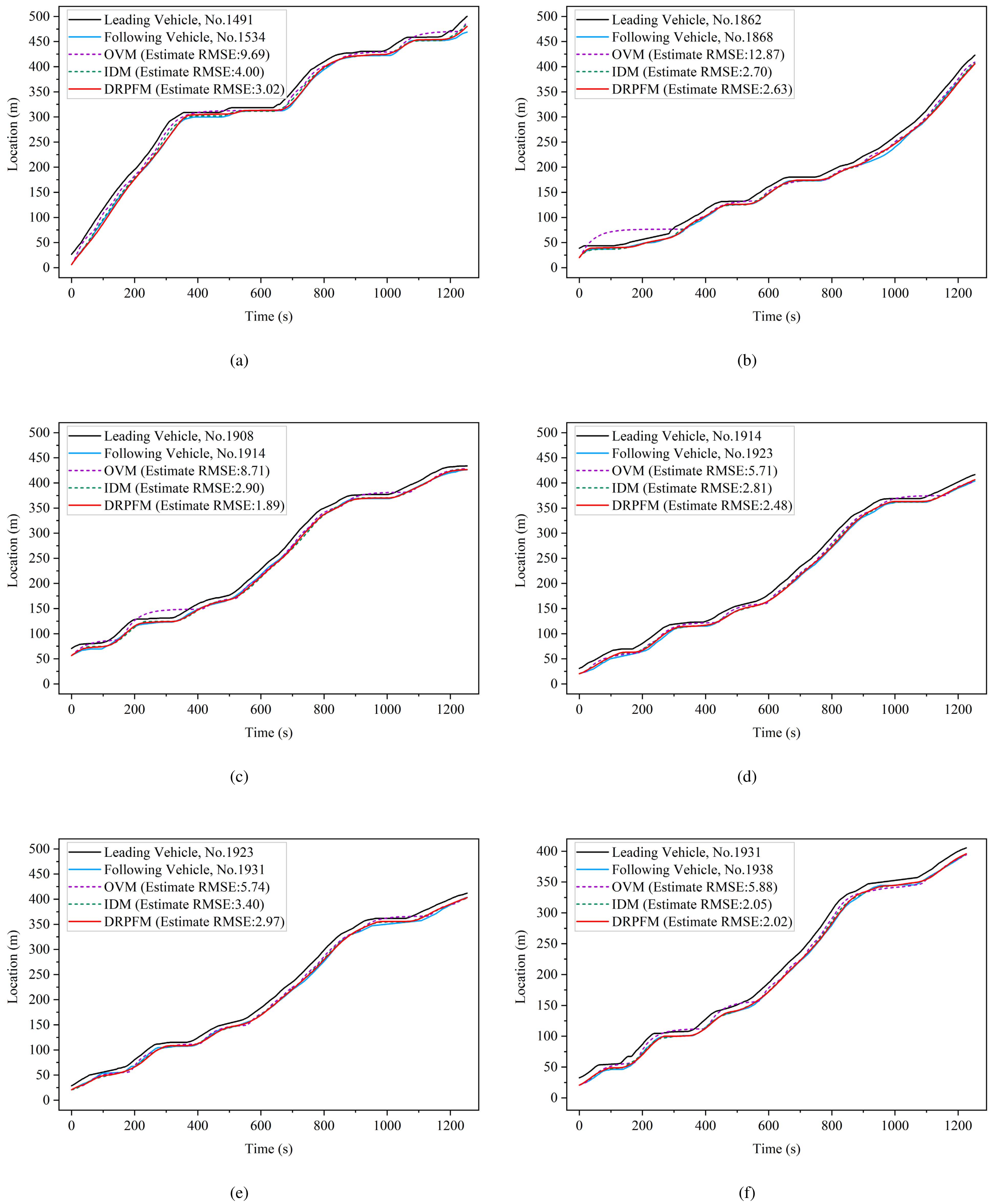


Fig. 13. Trajectory estimation performance by different car-following models.

Introducing the acceleration function of OVM into the safety criterion Eq. (17) with Eq. (18), we obtain the condition as follows:

$$\hat{s}_{\hat{f}} > s_{safe}^{OVM} = V^{-1}\left(\Delta x_{i,i-1}(t)\right)\left(v_{\hat{f}} - \tau' b_{safe}\right) \tag{21}$$

1) Rules for IDM

The expression of the IDM is shown in Eq. (22) - (23).

$$a^{IDM} = a_{\underset{\max}{IDM}}\left[1 - \left(\frac{v_i(t)}{v_{free}}\right)^{\delta^{IDM}} - \left(\frac{s^*}{s_\alpha}\right)^2\right] \tag{22}$$

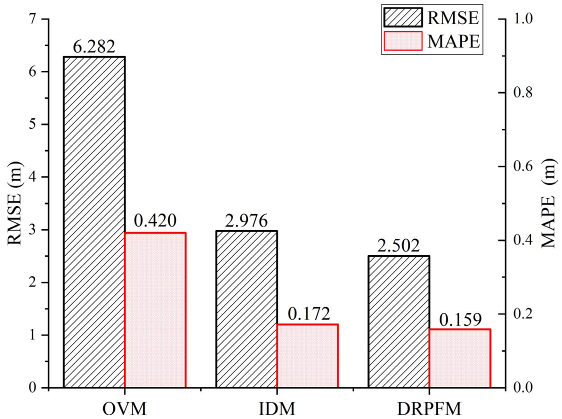


Fig. 14. Histogram of RMSE and MAPE for the OVM, IDM and DRPFM.

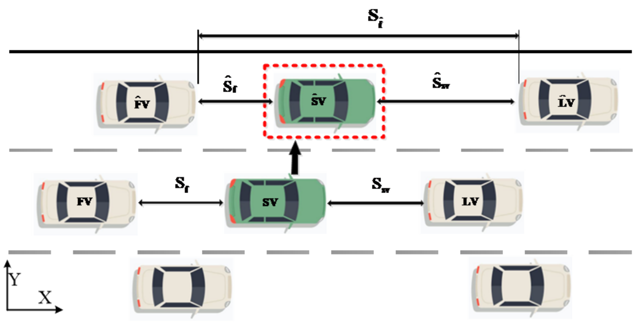


Fig. 15. Illustration of lane-changing scenarios.

$$s^* = s_0 + \max\left(0, vT + \frac{v\Delta v}{2\sqrt{ab}}\right) \tag{23}$$

The IDM safe gap function s_{safe}^{IDM} according to the safety criterion Eq. (17) can be analytically calculated through the following rules.

$$\hat{s}_{\hat{f}} > s_{safe}^{IDM} = \frac{s^*\left(v_{\hat{f}}, \Delta v\right)}{\sqrt{\frac{1}{a\left(a_{free}+b_{safe}\right)}}} \tag{24}$$

$$a_{free} = \lim_{\Delta x \to \infty} a\left(v, \Delta v, \Delta x\right) \tag{25}$$

1) Rules for DRPFM

The expression of the DRPFM is shown in Eq. (14).

Like OVM and IDM, we can also deduce the safe gap function s_{safe}^{DRPFM} of DRPFM. The results are as follows:

$$\hat{s}_{\hat{f}} > s_{safe}^{DRPFM} = \frac{M_a \lambda e^{\beta_1 a_1 + (\alpha+\beta_2)v_A + \beta_2 \Delta v}}{\sqrt{\frac{1}{\tau}\left(a_{free} + b_{safe}\right)}} \tag{26}$$

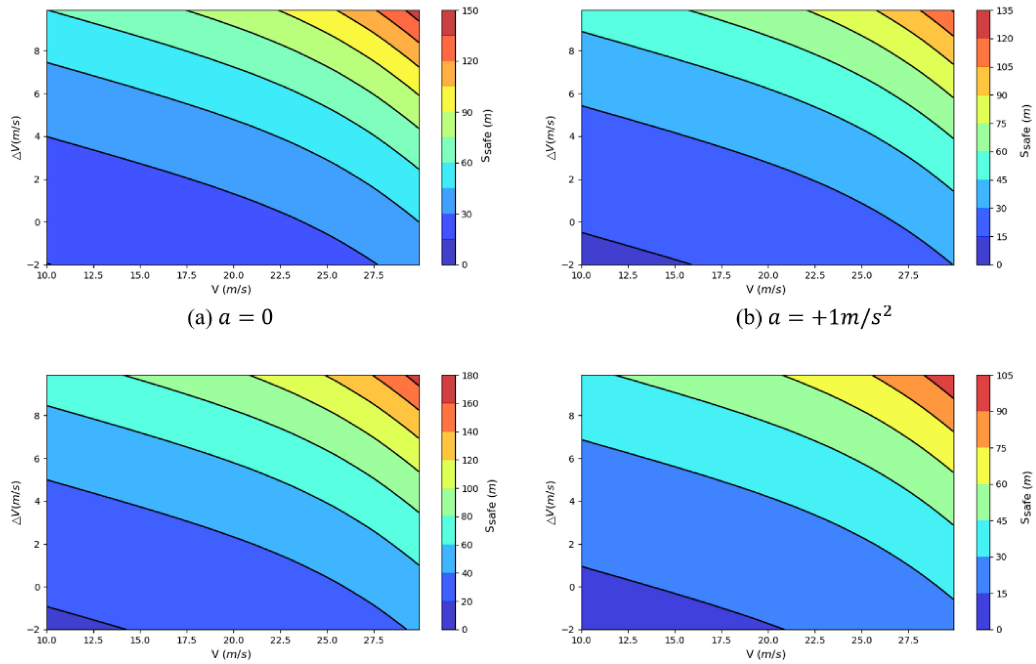
According to the description in Eq. (24), the minimum safety gap s_{safe} of the DRPFM is a function of the velocity of the lane-changing vehicle A v_A, and the speed difference $\Delta v = v_{\hat{f}} - v_A$ of the new follower with respect to the vehicle A under different acceleration values. The relationship between them is plotted below, as shown in Fig. 15.

Fig. 15 shows the minimum gap for our proposed DRPFM in Section III and$b_{safe} = 2m/s^2$. If, for example, in Fig. 15(a) both the changing vehicle A and the following vehicle in the target lane are driven at $25m/s$ (which means the $\Delta v = 0$) and $a = 0$ (the acceleration of vehicle A is zero), the minimum gap according to the safety criterion is about $30m$. At this gap, the new follower would have to brake with a deceleration b_{safe} in order to regain his or her safe gap. If, however, the new follower drives at 30 m/s, i.e., approaches by a rate of $\Delta v = 5$ m/s, the safety criterion displayed in Fig. 15(a) gives a minimum acceptable gap of about $45m$. As can be seen from Fig. 15(b) to Fig. 15(d), the minimum gap according to the safety criterion will change with the change of the value of vehicle acceleration. For example, in Fig. 15(b) and Fig. 15(d) $a = +1m/s^2$ and $+2m/s^2$, respectively, compared with Fig. 15(a), the minimum gap according to the safety criterion in Fig. 15(b) and Fig. 15(d) is smaller than the minimum gap in Fig. 15(a) when the values of v and δv are the same, and the minimum gap decreases with the increase of the value of a. However, by comparing Fig. 15(c) and Fig. 15(a), we can find that in Fig. 15(c), the acceleration is negative and the minimum gap according to the safety criterion in Fig. 15(c) is larger than the minimum gap in Fig. 15(a).

These phenomena above can be reasonably explained. The magnitude of the acceleration value during the lane-changing process can be used as the distinguishing standard of the drivers' type. When $a = 0$, drivers are normal drivers, a>0 drivers are aggressive drivers and a<0, drivers are timid drivers. Corresponding to Fig. 15, Fig. 15(a) represents normal drivers, Fig. 15(b) and Fig. 15(d) represent aggressive drivers and Fig. 15(c) represents timid drivers. The driving behavior of normal drivers is the most popular, representing the vast majority of drivers. Aggressive drivers tend to rely on overtaking and lane change to pursue speed gains, so the requirement for safe distance is not urgent, the minimum gap is usually small, and the minimum gap value decreases with the increase of acceleration. Timid drivers tend to decelerate and maintain a safe state, so they have strict requirements for safe distance, and the minimum gap is usually larger than for the other two types of drivers.

To summarise, we have the following findings:

1) driving risk potential field theory has a good performance with the modeling of microscopic traffic flow;
2) In the car-following scenario with obvious acceleration changes, the DRPFM has good performance in predicting the trajectory of the immediately following vehicle and has stronger performance than classical models (such as the OVM and IDM);
3) In the lane-changing scenario, the DRPFM can determine the minimum gap of different types of drivers at different velocities, accelerations and velocity differences.



(a) $a = 0$ (b) $a = +1m/s^2$

(c) $a = -1m/s^2$ (d) $a = +2m/s^2$

Fig. 16. Minimum safety Gap of the DRPFM under different acceleration.($\Delta v = v_{\hat{F}V} - v_{SV}$; and $v = v_{SV}$).

VII. Conclusion and Perspectives

In this study, we first propose an improved driving risk potential field to describe the safety risk in the CAVs environment. Compared with existing driving safety potential models, acceleration and the steering angle are incorporated into our model for the first time. This modified model of the driving risk potential field can reflect the dynamic distribution of the driving risk potential field under different speeds, accelerations and steering angles, which can be applied to describe the general traffic phenomena more naturally and can be used to analyze the motion characteristics of the vehicle's microscopic level. Additionally this driving risk potential field model is applied to the microscopic modeling process of the traffic flow. A car-following model based on this driving risk potential field model has been developed, and an analysis has been made of the safety criterion for lane change behavior.

The contribution to the academic includes two parts, one is the driving risk potential field modeling research, and the other is the traffic flow modeling research. Dynamic driving risk potential field model proposed in this paper is a big improvement over conventional models, which incorporates the acceleration information of vehicles.Our work has achieved the transition from a relatively static potential field model to a dynamic potential field model. Compared with existing potential field models, this dynamic model is much more suitable for the research of traffic problems in the connected and automated environment. With respect to the car-following model established in this paper, it can be used as an alternative car-following model under the automated and connected environment in the future. In addition, this modeling method based on the driving risk potential filed can also be applied to the lane-change modeling, and macroscopic traffic modeling.

From the perspective of traffic safety control technology in a connected and automated environment, safer driving has always been the research direction we are pursuing. The findings in this study can more accurately assess the degree of vehicle driving risk and thus lay a new foundation for investigation of intelligent driver assistance systems, such as vehicle safety assistance and intelligent control.

This study reveals the great potential for applying new approaches to the driving risk potential field.There are some limitations of this study that would need further improvements. First, more available field data needs to be collected for the purpose of calibration and validation of our proposed model. Second, the possibilities of application in other traffic problem areas need to be investigated in the future. Therefore, further study may focus on the application of the dynamic driving risk potential model in dealing with various traffic problems, e.g., macroscopic traffic flow modeling, dynamic path planning, automatic control method of autonomous vehicles.Nevertheless, we believe our work in this paper may open a door towards alternative to conventional traffic flow models and vehicle control method for many tasks.

Appendix

Description of Three Basic Principles:

1. The acceleration is a strictly decreasing function of the velocity. Moreover, the vehicle accelerates towards a desired velocity v_0 if not constrained by other vehicles or obstacles:

$$\frac{\partial a(\Delta x, v, \Delta v)}{\partial v} < 0$$

2. The acceleration is an increasing function of the distance δx to the leading vehicle:

$$\frac{\partial a(\Delta x, \upsilon, \Delta \upsilon)}{\partial \Delta x} \geq 0$$

3. The acceleration is an increasing function of the velocity of the leading vehicle. Together with requirement (1), this also means that the acceleration decreases (the deceleration increases) with the velocity of approach to the lead vehicle (or obstacle):

$$\frac{\partial a(\Delta x, \upsilon, \Delta \upsilon)}{\partial \Delta \upsilon} \leq 0$$

Proof: For the DRPFM, we obtain the derivatives of υ_B, Δx and $\Delta \upsilon$ as following:

$$\dot{\upsilon_B} = a_{max} \tanh\left[\delta \cdot \left(\upsilon_0^{(\alpha)} - \upsilon_B\right)\right] - M_A \lambda \frac{e^{-\beta_1 a_A + \alpha \upsilon_A}}{\tau \cdot \Delta x} \cdot e^{\beta_2 \upsilon_B}$$

For the requirement (1), derive the velocity υ_B from the acceleration equation:

$$\frac{\partial \dot{\upsilon}_B}{\partial \upsilon_B} = -a_{max} sech^2\left[\delta\left(\upsilon_0^{(\alpha)} - \upsilon_B\right)\right] \cdot \delta - M_A \lambda \frac{e^{-\beta_1 a_A + \alpha \upsilon_A + \beta_2 \upsilon_B}}{\tau \Delta x} \cdot \beta_2 < 0$$

For the requirement (2), derive the distance Δx from acceleration equation:

$$\frac{\partial \dot{\upsilon}_B}{\partial \upsilon_B} = M_A \lambda \frac{e^{-\beta_1 a_A + \alpha \upsilon_A + \beta_2 \upsilon_B}}{\tau} \cdot \frac{1}{\Delta^2 x} > 0$$

For the requirement (3), derive the velocity of approach to the lead vehicle $\Delta \upsilon$ from acceleration equation:

$$\frac{\partial \dot{\upsilon}_B}{\partial \upsilon_B} = -M_A \lambda \frac{e^{-\beta_1 a_A + (\alpha + \beta_2) \upsilon_A + \beta_2 \Delta \upsilon}}{\tau \Delta x} \cdot \beta_2 < 0$$

Thus DRPFM satisfy these three basic requirements, which completes the proof.■

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